

VPC USING BASTION SERVER

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1. VPC with Bastion Server

VPC & Subnet Design

VPC CIDR: 10.0.0.0/16

The screenshot displays the AWS Management Console interface for creating a new VPC. The 'Create VPC' page is titled 'Create VPC' and includes a brief description: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.' The 'VPC settings' section is the primary focus, containing several configuration options. Under 'Resources to create', the 'VPC only' radio button is selected. The 'Name tag - optional' field is populated with 'MyVPC'. In the 'IPv4 CIDR block' section, 'IPv4 CIDR manual input' is chosen, and the 'IPv4 CIDR' field contains '10.0.0.0/16'. The 'IPv6 CIDR block' section has 'No IPv6 CIDR block' selected. The 'Tenancy' dropdown is set to 'Default'. The 'VPC encryption control (B)' section shows 'None' as the selected option. Below this, the 'Tags' section allows for adding a tag with the key 'Name' and the value 'MyVPC'. At the bottom right of the form, there are three buttons: 'Cancel', 'Preview code', and 'Create VPC'.

Public Subnet (Bastion Server): 10.0.1.0/24

us-east-1.console.aws.amazon.com / VPC us-east-1

Create subnet

VPC
VPC ID: vpc-df39ca22a1564798 (MyVPC)
Associated VPC CIDRs: 10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Public Subnet

Availability Zone
United States (N. Virginia) / us-east-1a (us-east-1a)

IPv4 VPC CIDR block
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24

Tags - optional

Key	Value - optional
Name	Public Subnet

[Add new tag](#)
[Remove](#)
[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Private Subnet (Database Server): 10.0.2.0/24

us-east-1.console.aws.amazon.com / VPC us-east-1

Create subnet

VPC
VPC ID: vpc-df39ca22a1564798 (MyVPC)
Associated VPC CIDRs: 10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
PrivateSubnet

Availability Zone
No preference

IPv4 VPC CIDR block
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24

Tags - optional

Key	Value - optional
Name	PrivateSubnet

[Add new tag](#)
[Remove](#)
[Add new subnet](#)

[Cancel](#) [Create subnet](#)

2. Security Group Configuration (VERY IMPORTANT)

Bastion Server Security Group (Public Subnet)

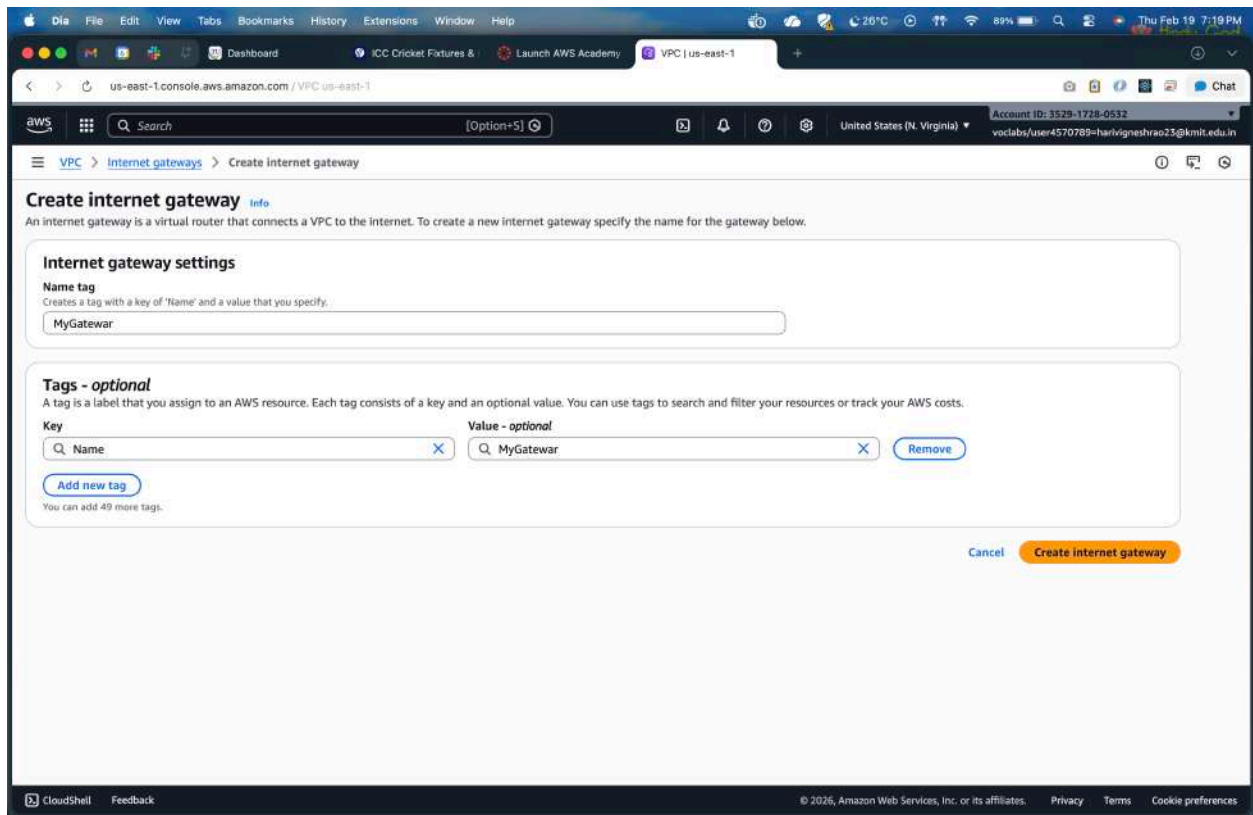
Allows SSH (22) from: My IP (Database Administrator)

Database Server Security Group (Private Subnet)

Allows MySQL / Aurora (3306) from: custom 10.0.1.0/24 (Public subnet / Bastion subnet)

Allows SSH (22) from: custom Private IP of Bastion Server

Does NOT allow public access



us-east-1 console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Route tables > Create route table

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

PuliRoute

VPC
The VPC to use for this route table.

vpc-0fc39ca22a1564798 (MyVPC)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Name

Value - optional

PuliRoute

Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

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us-east-1 console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Internet gateways > Attach to VPC (igw-0403ca9fa3033e631)

Attach to VPC (igw-0403ca9fa3033e631)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-0fc39ca22a1564798

AWS Command Line Interface command

Cancel Attach internet gateway

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us-east-1 console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Route tables > rtb-0888270d42fc59abf > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute
	igw-0403ca9fa3033e631			

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

us-east-1 console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Route tables > rtb-0888270d42fc59abf > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public Subnet	subnet-064eb229c0822dc3b	10.0.1.0/24	-	Main (rtb-0f38df46682081c92)
☐	PrivateSubnet	subnet-0d046f4b1f405849e	10.0.2.0/24	-	Main (rtb-0f38df46682081c92)

Selected subnets

subnet-064eb229c0822dc3b / Public Subnet

[Cancel](#) [Save associations](#)

us-east-1.console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Route tables > Create route table

Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

PrivateRoute

VPC

The VPC to use for this route table.

vpc-0fc39ca22a1564798 (MyVPC)

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key

Q Name X

Value - optional

Q PrivateRoute X Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

CloudShell

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us-east-1.console.aws.amazon.com / VPC us-east-1

Account ID: 3529-1728-0532
voclabs/user4570789-harivigneshrao23@kmit.edu.in

VPC > Route tables > rtb-0041b67cf2fe2809d > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	Public Subnet	subnet-064eb229c0822dc3b	10.0.1.0/24	-	rtb-0888270d42fc59abf / PullRoute
☒	PrivateSubnet	subnet-0d046f4b1f405849e	10.0.2.0/24	-	Main (rtb-0f38df46682081c92)

Selected subnets

subnet-0d046f4b1f405849e / PrivateSubnet X

Cancel

Save associations

CloudShell

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us-east-1.console.aws.amazon.com / VPC us-east-1

Create NAT gateway

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
NATgateway
The name can be up to 256 characters long.

Availability mode
Choose whether to deploy across all zones in the region or restrict to a single availability zone.
☒ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi-AZ deployments.
☐ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet-level settings.

VPC
Select a VPC in which to create the regional NAT gateway.
vpc-0fc39ca22a1564798 (MyVPC)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ Private

Method of Elastic IP (EIP) allocation
Choose how IP addresses are associated with NAT gateways.
☒ **Automatic**
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling—adding AZs automatically allocates EIPs, simplifying management.
☐ **Manual**
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.
Key
Q Name
Value - optional
Q NATgateway
Add new tag
You can add 49 more tags.

Cancel Create NAT gateway

us-east-1.console.aws.amazon.com / VPC us-east-1

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local	-	No	CreateRoute
	Q NAT Gateway			
	Q nat-19404ad72892afdc5			

Add route

Cancel Preview Save changes

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name:

Description:

VPC:

Inbound rules

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	My IP 183.82.235.53/32	

Outbound rules

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom 0.0.0.0/0	

Rules with destination of 0.0.0.0/0 or ::0/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

No tags associated with the resource.

[Add new tag](#)

[Cancel](#) [Create security group](#)

3. Security Outcome

Database server is NOT publicly accessible

Only Bastion server can connect to DB server using SSH

External users cannot access DB directly

4. EC2 Instance Creation

Bastion Server

Subnet: Public Subnet (10.0.1.0/24)

Key pair: bastion.pem

Operating System: Ubuntu

Database Server

Subnet: Private Subnet (10.0.2.0/24)

Key pair: dbserver.pem

Operating System: Ubuntu

aws

Search

[Option+5]

United States (N. Virginia)

Account ID: 3129-1728-0532

voclabs/user4570789-harivigneshrao23@kmit.edu.in

EC2

Instances

Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

BastionServer

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0b6c5ebd2801a5cb (64-bit (x86)) / ami-0071e8c431ea0eddb (64-bit (ARM))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

CloudShell Feedback

Summary

Number of instances [Info](#)

1

Software image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-0b6c5ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

aws

Search

[Option+5]

United States (N. Virginia)

Account ID: 3129-1728-0532

voclabs/user4570789-harivigneshrao23@kmit.edu.in

EC2

Instances

Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

bastion

Create new key pair

Network settings

VPC - required

vpc-0fc39ca22a1564798 (MyVPC)

Subnet

subnet-064eb229c0822dc3b

Public Subnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

BastionSecurity sg-0bc0bf7ad7167b240

Common security groups

Select security groups

Compare security group rules

Advanced network configuration

Summary

Number of instances [Info](#)

1

Software image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)

ami-0b6c5ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

BastionSecurity

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

[Preview code](#)

EC2 > Security Groups > Create security group

Create security group

Basic details

Security group name

DatabaseSecurity

Description

allowssh

VPC

vpc-df53ba22a1564798 (MyVPC)

Inbound rules

Type

SSH

Protocol

TCP

Port range

22

Source

Custom

Description - optional

MySQL/Aurora

TCP

3306

Custom

10.0.1.0/24

10.0.1.0/24

Add rule

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Add rule

Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Create security group

EC2 > Instances > Launch an instance

Launch an instance

Name and tags

Name

DatabaseServer

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.

Search our full catalog including 1000s of application and OS images.

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0b6c6ebd2801a5cb / ami-0071c8c431eaa0db (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0b6c6ebd2801a5cb

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

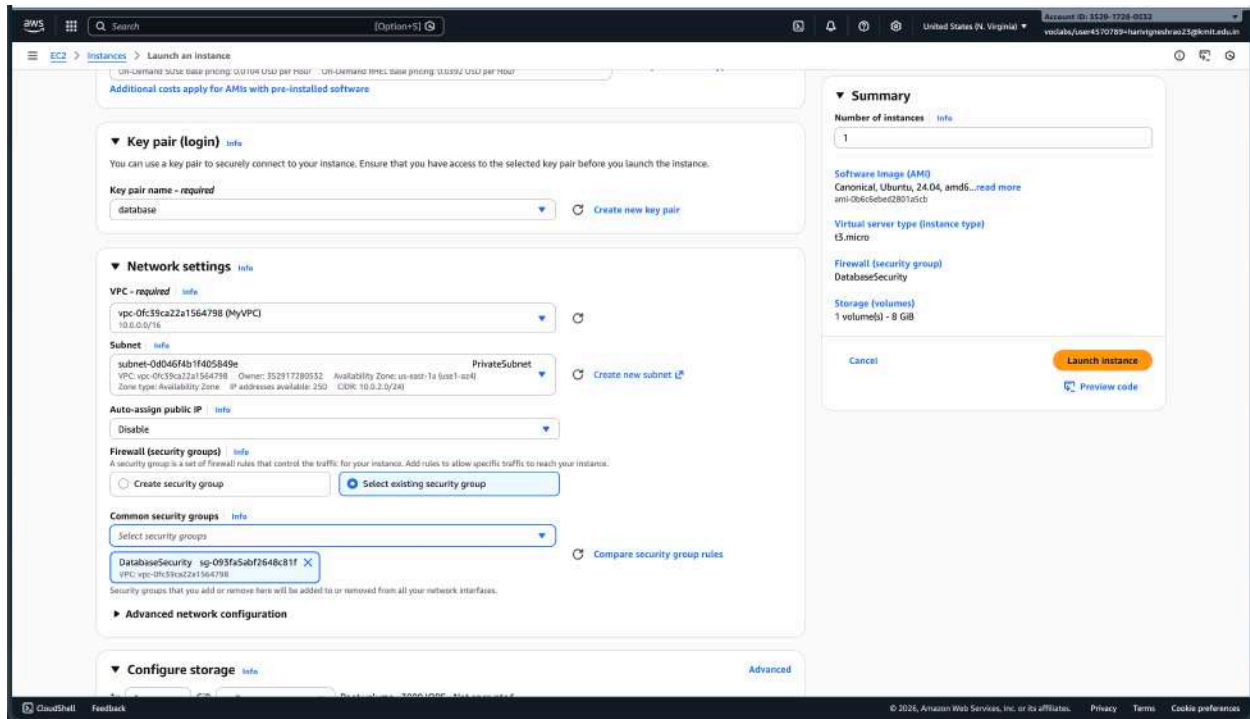
Cancel

Launch instance

Preview code

CloudShell Feedback

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5. Copy DB Server Key to Bastion Server

Database server is in private subnet

Direct SSH from local system is NOT possible

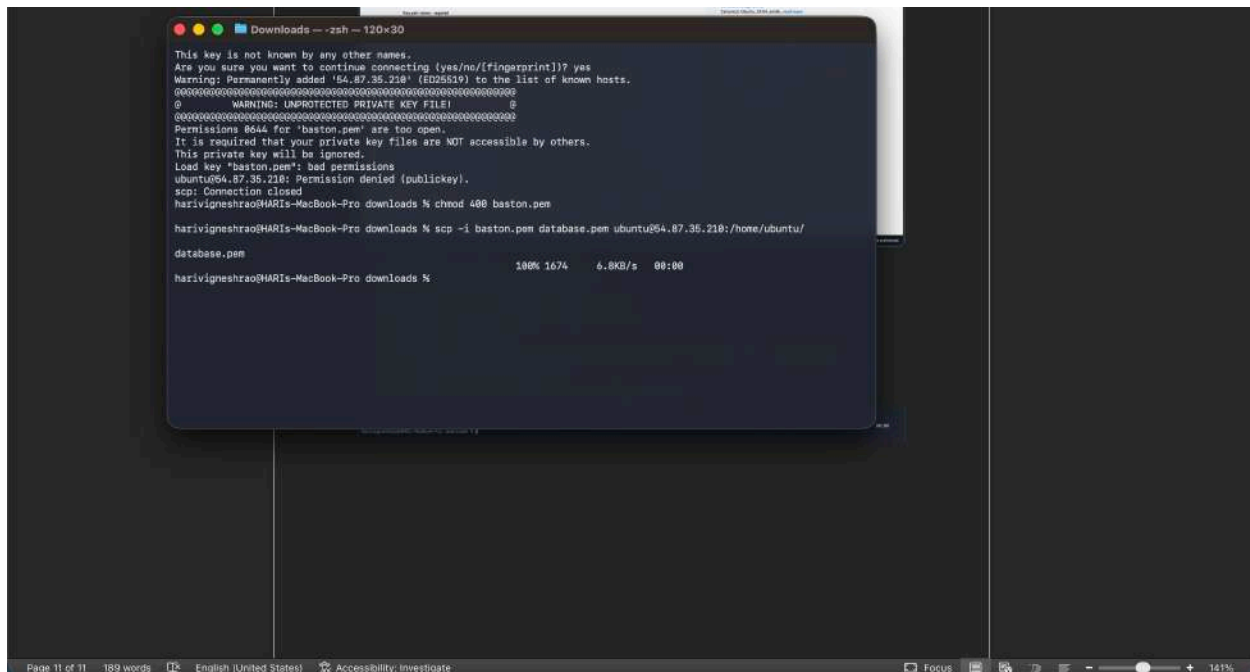
DB server key must be copied to Bastion server

Command (PowerShell):

```
scp -i bastion.pem dbserver.pem ubuntu@<BASTION_PUBLIC_IP>:/home/ubuntu/
```

File is copied securely using SCP

dbserver.pem will now be available inside Bastion server

A terminal window titled "Downloads -- zsh -- 120x30" is shown. It displays the output of an SSH command. The output includes a warning about a new host key, a confirmation to add it, a warning about an unprotected private key file, and a message that the private key will be ignored. It then shows the loading of the key, a permission error for the bastion.pem file, and the successful connection to the bastion server. The user then runs 'chmod 400 bastion.pem' and 'scp -i bastion.pem database.pem ubuntu@54.87.35.218:/home/ubuntu/database.pem'. The scp command shows a progress bar at 100% and a transfer rate of 6.8KB/s. The terminal window is part of a larger application interface with a dark background and a status bar at the bottom showing "Page 11 of 11", "189 words", "English (United States)", "Accessibility: Investigate", "Focus", and "141%".

```
Downloads -- zsh -- 120x30
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.87.35.218' (ED25519) to the list of known hosts.
@
WARNING: UNPROTECTED PRIVATE KEY FILE!
@
Permissions 0644 for 'bastion.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key 'bastion.pem': bad permissions
ubuntu@54.87.35.218: Permission denied (publickey).
scp: Connection closed
harivigneshrao@HARIS-MacBook-Pro downloads % chmod 400 bastion.pem
harivigneshrao@HARIS-MacBook-Pro downloads % scp -i bastion.pem database.pem ubuntu@54.87.35.218:/home/ubuntu/
database.pem
harivigneshrao@HARIS-MacBook-Pro downloads %
100% 1674 6.8KB/s 00:00
```

6. Connect to Bastion Server

SSH Command:

```
ssh -i bastion.pem ubuntu@<BASTION_PUBLIC_IP>
```

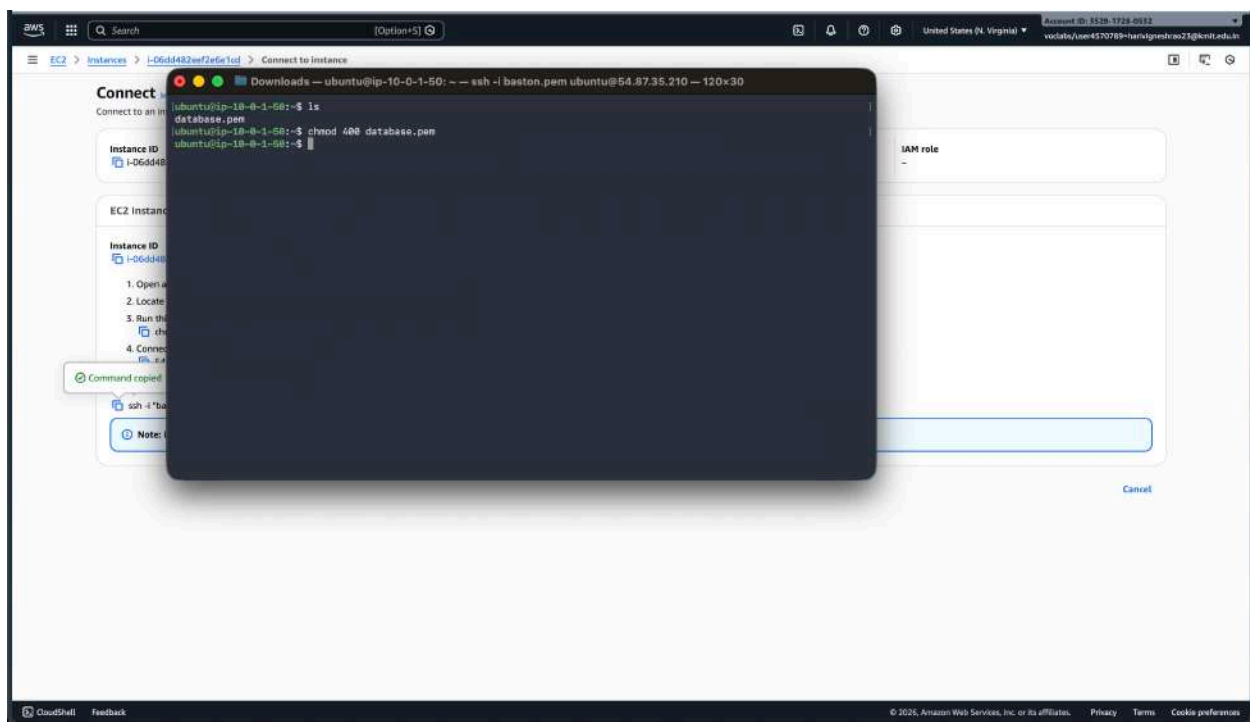
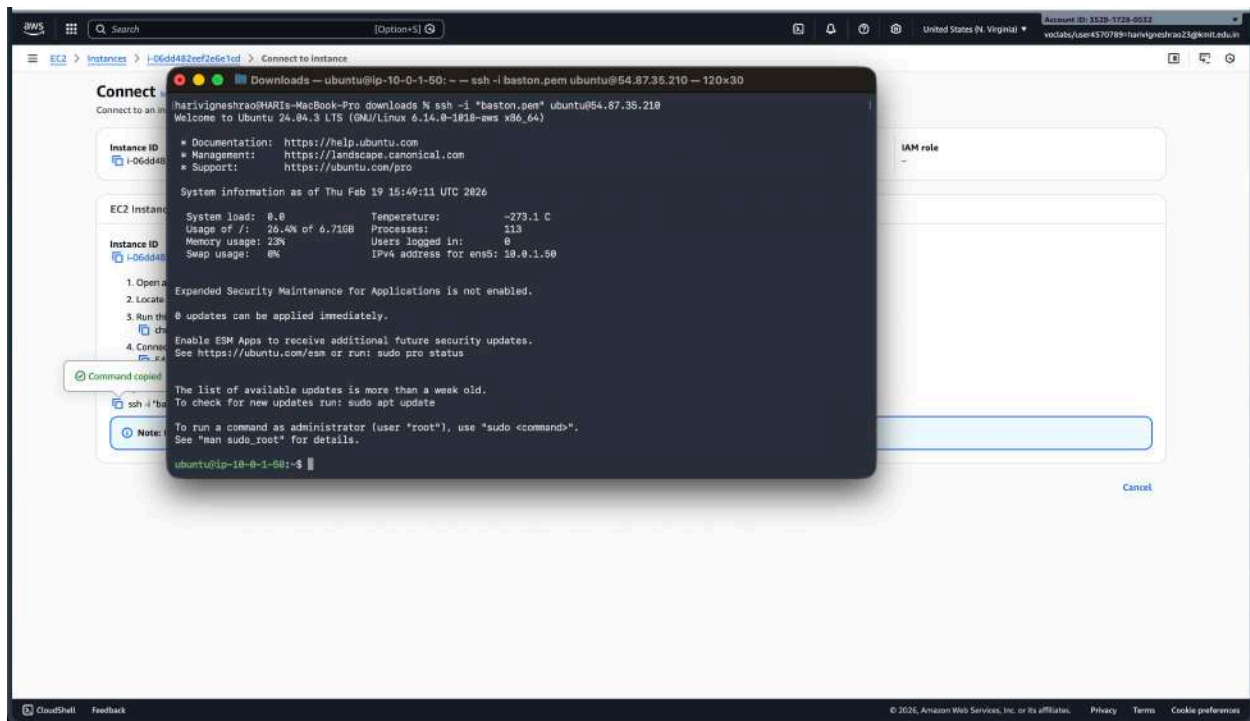
Verify key files:

```
ls
```

Output:

```
bastion.pem
```

```
dbserver.pem
```



7. Connect from Bastion to Database Server

SSH from Bastion:

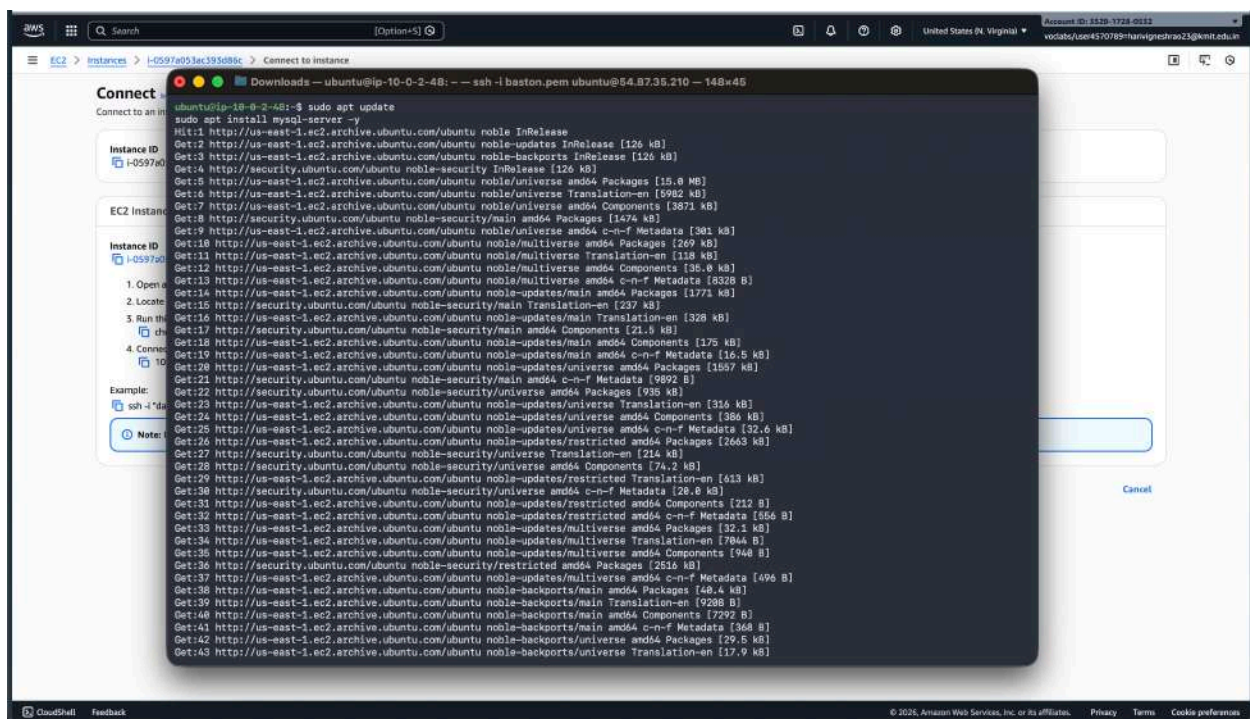
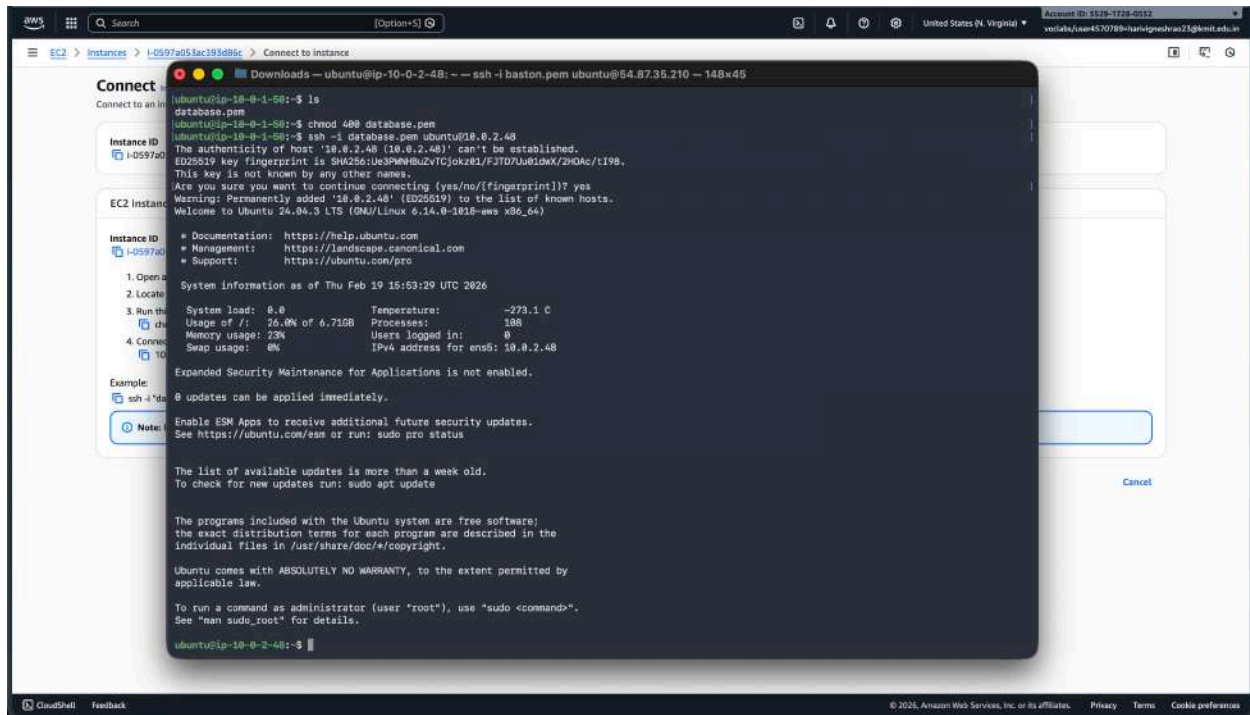
```
ssh -i dbserver.pem ubuntu@10.0.2.x
```

Connection to DB server is successful

Database administrator can now:

Update database configurations

Install required packages-----problem occurs Restart MySQL service



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Account ID: 1128-1728-0132

vp1abv/iam4570789-harishgopalvan25@kmit.edu.in

United States (N. Virginia)

[Option+5]

EC2 > Instances > i-0597a053ac393086c > Connect to instance

Connect to an instance

Instance ID: i-0597a053

EC2 Instance Profile: AmazonEC2RoleforSSM

1. Open a terminal window

2. Locate the instance profile

3. Run the command

4. Connect to the instance

Example: ssh -i "path-to-key-pair.pem" ubuntu@ip-10-0-2-48

Note: The instance profile is used to grant the instance access to the AWS services.

Downloads — ubuntu@ip-10-0-2-48: ~ — ssh -i baston.pem ubuntu@54.87.35.210 — 148x45

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

ubuntu@ip-10-0-2-48:~\$ sudo systemctl start mysql

sudo systemctl enable mysql

Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.

Executing: /usr/lib/systemd/systemd-sysv-install enable mysql

ubuntu@ip-10-0-2-48:~\$ sudo systemctl status mysql

● mysql.service - MySQL Community Server

Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)

Active: active (running) since Thu 2025-02-19 15:55:11 UTC; 59s ago

Main PID: 2223 (mysqld)

Status: "Server is operational"

Tasks: 38 (limit: 1088)

Memory: 351.3M (peak: 377.9M)

CPU: 1.331s

CGroup: /system.slice/mysql.service

└─2223 /usr/sbin/mysqld

Feb 19 15:55:18 ip-10-0-2-48 systemd[1]: Starting mysql.service - MySQL Community Server...

Feb 19 15:55:11 ip-10-0-2-48 systemd[1]: Started mysql.service - MySQL Community Server.

ubuntu@ip-10-0-2-48:~\$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords and improve security. It checks the strength of password and allows the users to set only those passwords which are secure enough. Would you like to setup VALIDATE PASSWORD component?

!Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW Length >= 8

MEDIUM Length >= 8, numeric, mixed case, and special characters

STRONG Length >= 8, numeric, mixed case, special characters and dictionary

file

Cancel